Cheatsheet: Interactive Code with Pyodide

Running Python live in the browser

1 — Install & enable

> ⚠ Interactive in HTML only
>
> Running and editing works only in **HTML output** (`format: html`), that is in websites and books. In PDF or Word the same code appears as ordinary code that cannot be run — see the Exercise Sheet cheatsheet for code that has to work everywhere.

`pyodide-interaktiv` is a rework of `coatless-quarto/pyodide` that adds real `input()`, animations, a Stop button, and AI feedback. It installs like any other extension:

Listing 1 `Terminal`

```
quarto add Erasmus-CTM/pyodide-interaktiv
```

Enable it in a page's YAML header:

```yaml
---
title: "My Page"
format: html
filters:
  - Erasmus-CTM/pyodide-interaktiv
---
```

2 — Writing interactive cells

Use a fence tagged `{pyodide-python}` instead of a regular code block:

````
```{pyodide-python}
x = [1, 2, 3, 4, 5]
print(sum(x))
```
````

Students can edit this code directly in the browser and run it with the **Run code** button (or `Shift + Enter`) — nothing to install on their end.

Check it: run `quarto preview`. The cell appears with a row of buttons. On the first click on **Run code** Python loads, which takes a few seconds.

> Want graded exercises instead?
>
> `py-exercise` adds code blocks with hidden automated tests; `math-exercise` checks symbolic/numeric answers with SymPy. Both are installed the same way (`quarto add Erasmus-CTM/py-exercise`, `quarto add Erasmus-CTM/math-exercise`) — live examples are on the Example Page of this tutorial.

3 — What pyodide-interaktiv gives you

Python runs in a **Web Worker**, that is next to the page rather than inside it. As a result:

Feature	What it means
Real `input()`	A cell can pause and wait for typed input while the page stays usable
Animations	A Matplotlib `FuncAnimation` plays as an interactive player
Safe infinite loops	An accidental `while True:` does not lock up the page — **Stop** cancels it
AI feedback	Every cell has a **Feedback** button (needs to be set up first); the hints get more concrete with each click
Responsive page	Long calculations don't freeze the page
Shared variables	All cells on a page belong to one Python session. A later cell can use what an earlier one defined or imported, once that one has been run

Try `input()` and shared variables live on the Code page, and the animation on the Visualizations page of the Example Page.

Full details: Interactive Python with Pyodide